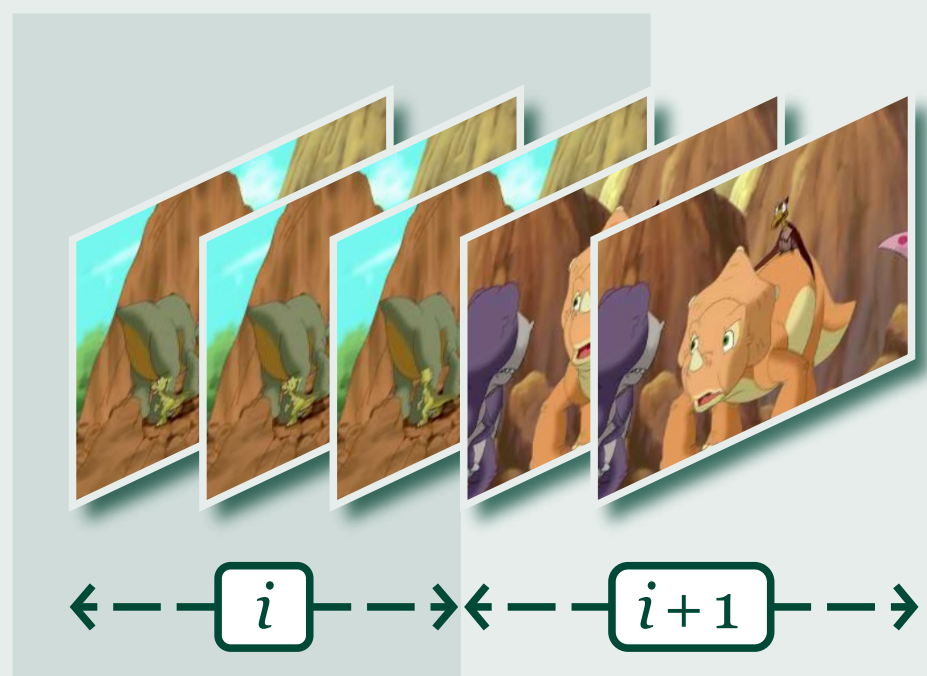




Shot segmentation



Character detection



Multi-object tracking



Contrastive fine-tuning

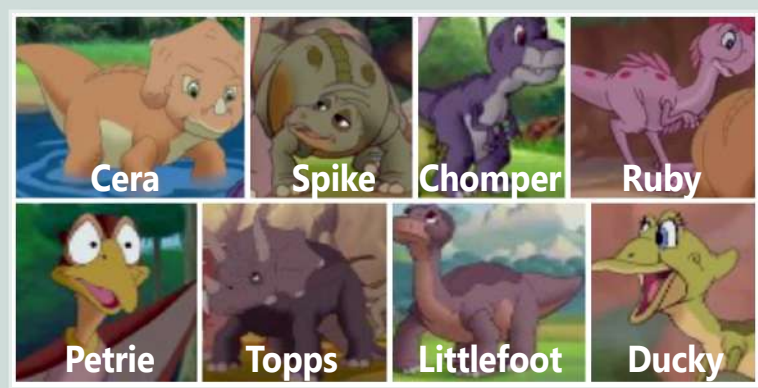


Semantic embedding

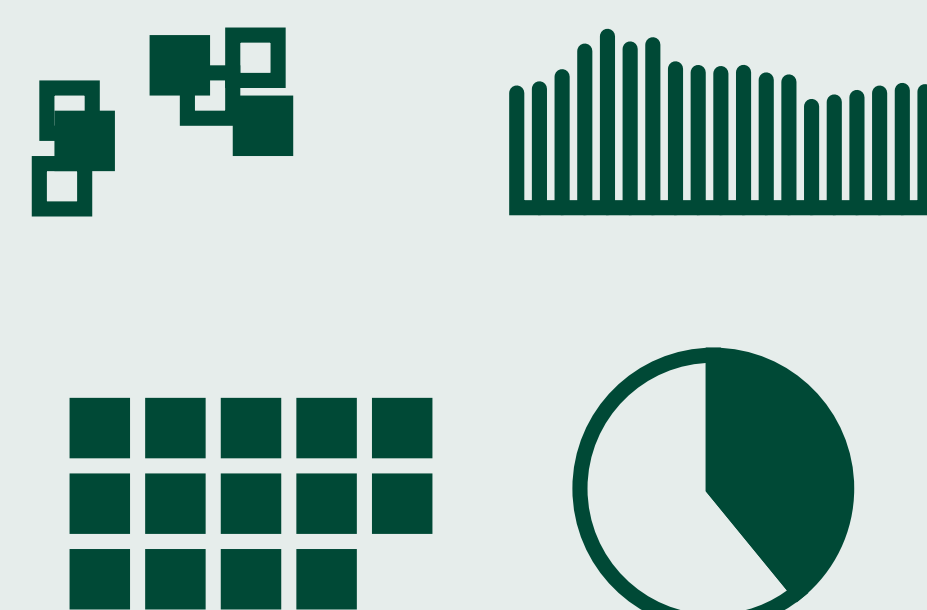
Candidates naming



Training set



Multi-class character classifier



Visualizations

Visualizing *the* C.A.S.T. of Characters Labeled in Animated Videos

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The success of a visualization lies not only in the design of the depiction, but often also on the underlying representation used for the data. In a recent paper [1] we presented a method called CAST: Character labeling in Animation using Self-supervision by Tracking. This method learns a semantic representation suitable for character identification and labeling specific to a given animation style using self supervision.

Recognizing and labeling characters in a given video is an important task for many applications. It can empower media companies to search, analyze, visualize and reuse content in videos.

However, relying on current computer vision and deep-learning solutions often fails these models were trained on natural images/videos, that have very different appearance characteristics compared to cartoons and drawn animations. In addition, there is a wide diversity of styles and genres in animation such as Manga, hand-drawn, and CGI movies that can have significant differences in appearance.

Lastly, characters in animation videos could be anything from a person to an animal or even an object like a talking candelabra, and the same character can change color, texture, style, and shape, even in the very same scene. Such variability means that the semantic representation of a character cannot be based on appearance alone, cannot use natural images learned representations and must include several possible instance variations of the same character.

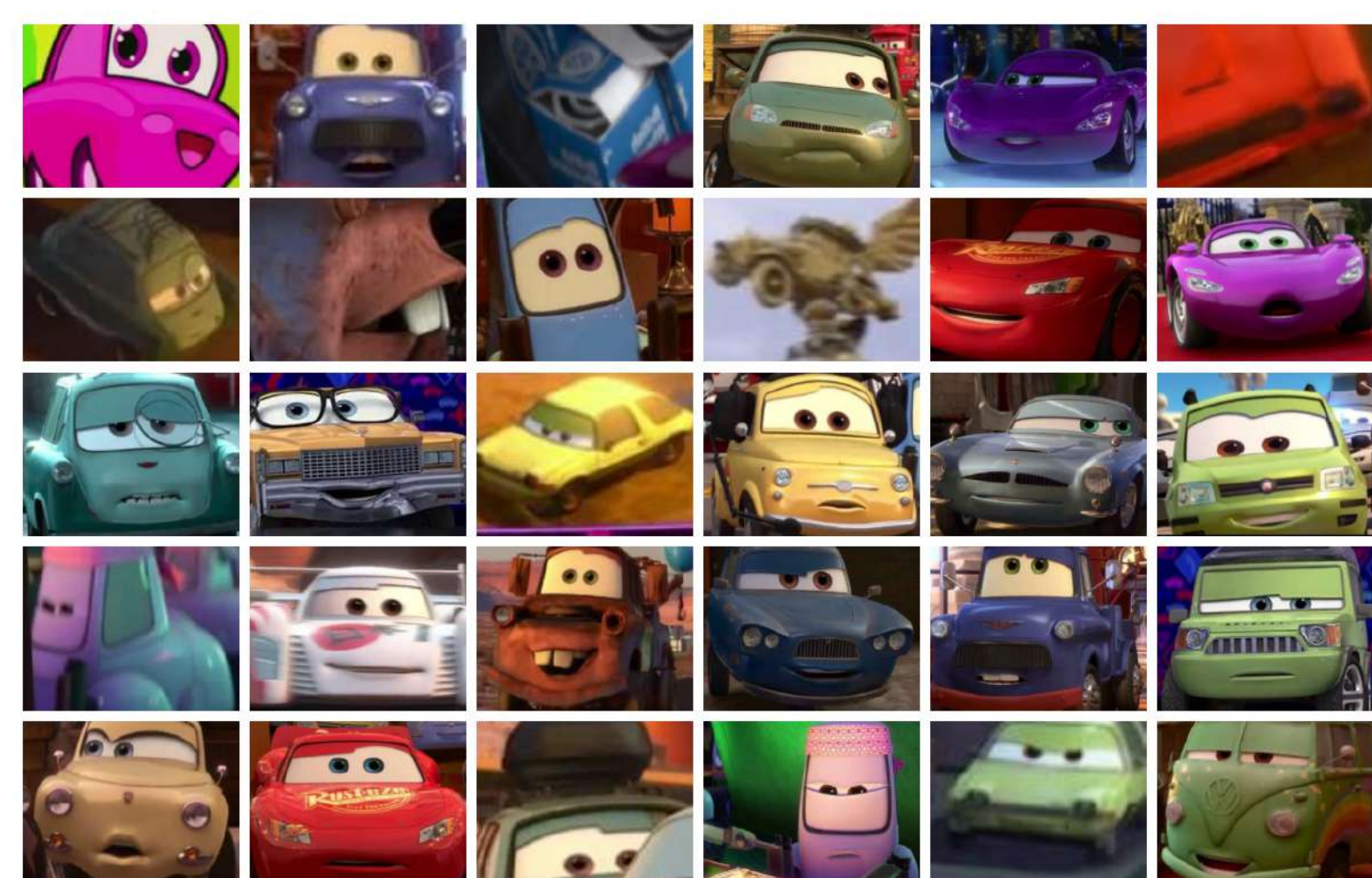


Figure 2:
An example of a dictionary output on 'Cars 2'. Note how five similar green cars were identified correctly as different characters

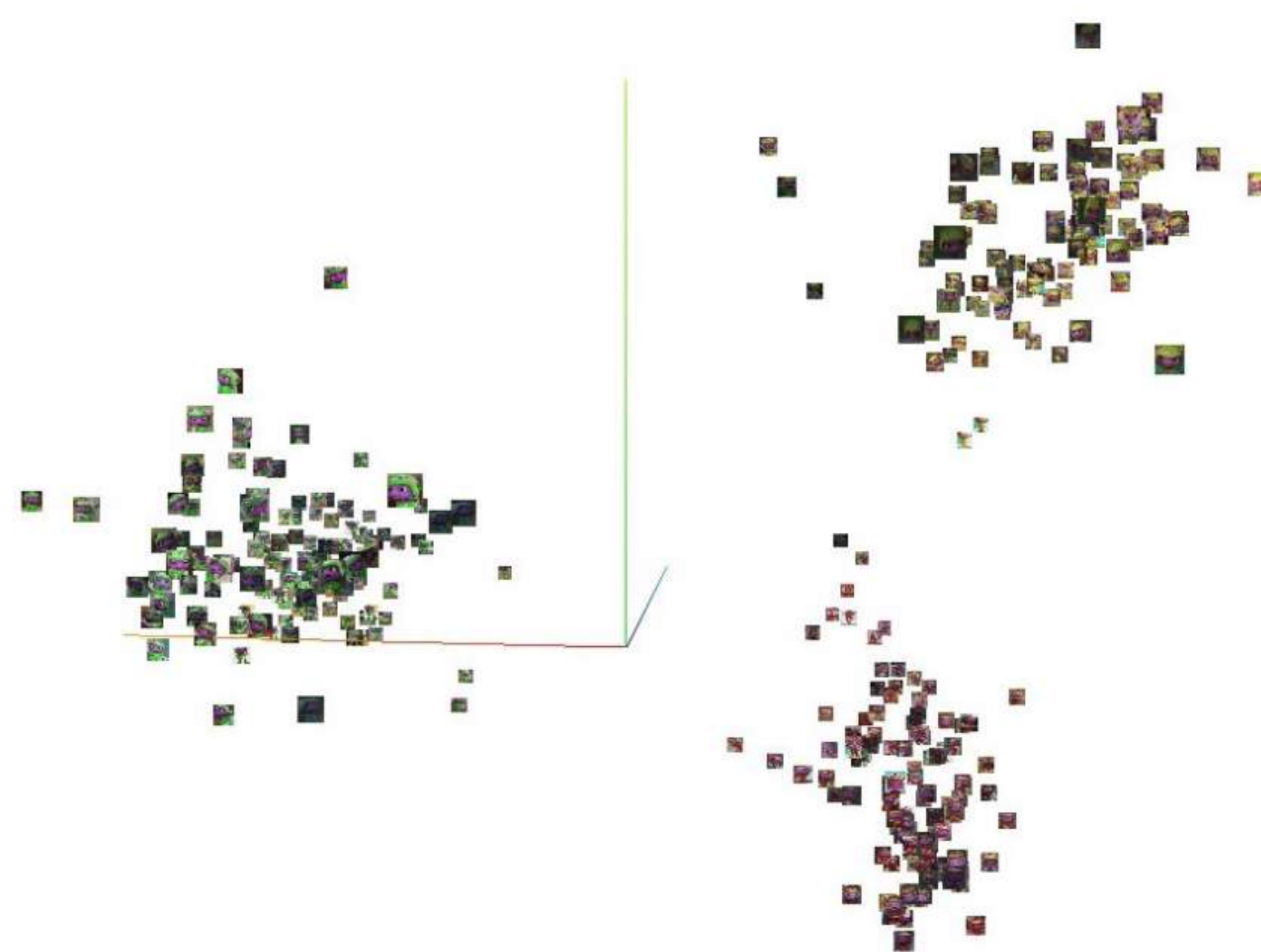


Figure 4:
A projection onto 3D space of the semantic embedding creates a structured navigation space for all characters' depictions in the video.



Figure 6:
Analyzing the movement of characters within a shot.

An overview of the method can be seen in Figure 1 (left). As a first step, we segment the input video to shots, and run a base object detector that was refined for detection using a labeled dataset of multiple animated styles. We follow each detection bounding box in the shot through changes in appearance and shape using a multi-object tracker. To adapt the representation towards the specific animation style and move different instances of the same character closer in the semantic space, we further refine the base detector using contrastive learning.

The underlying hypothesis is that two different detections which appear in the same frame do not share identity, while two detections within the same tracklet do. This assumption allows us to automatically sample and create a large number of triplet examples for contrastive training with triplet loss. The refinement adapts the semantic mapping towards representing the specific animated style, where distances convey the identity of characters and not simply appearance differences. CAST was tested on numerous animation videos in various styles.

Using the style-specific representation we first built a dictionary of characters in a video. Second, we defined highly accurate classifiers that can be used to densely identify and label characters. Using these tools along with the underlying semantic embedding space allows us to analyze and visualize numerous video-level and character level attributes and trends as shown next.

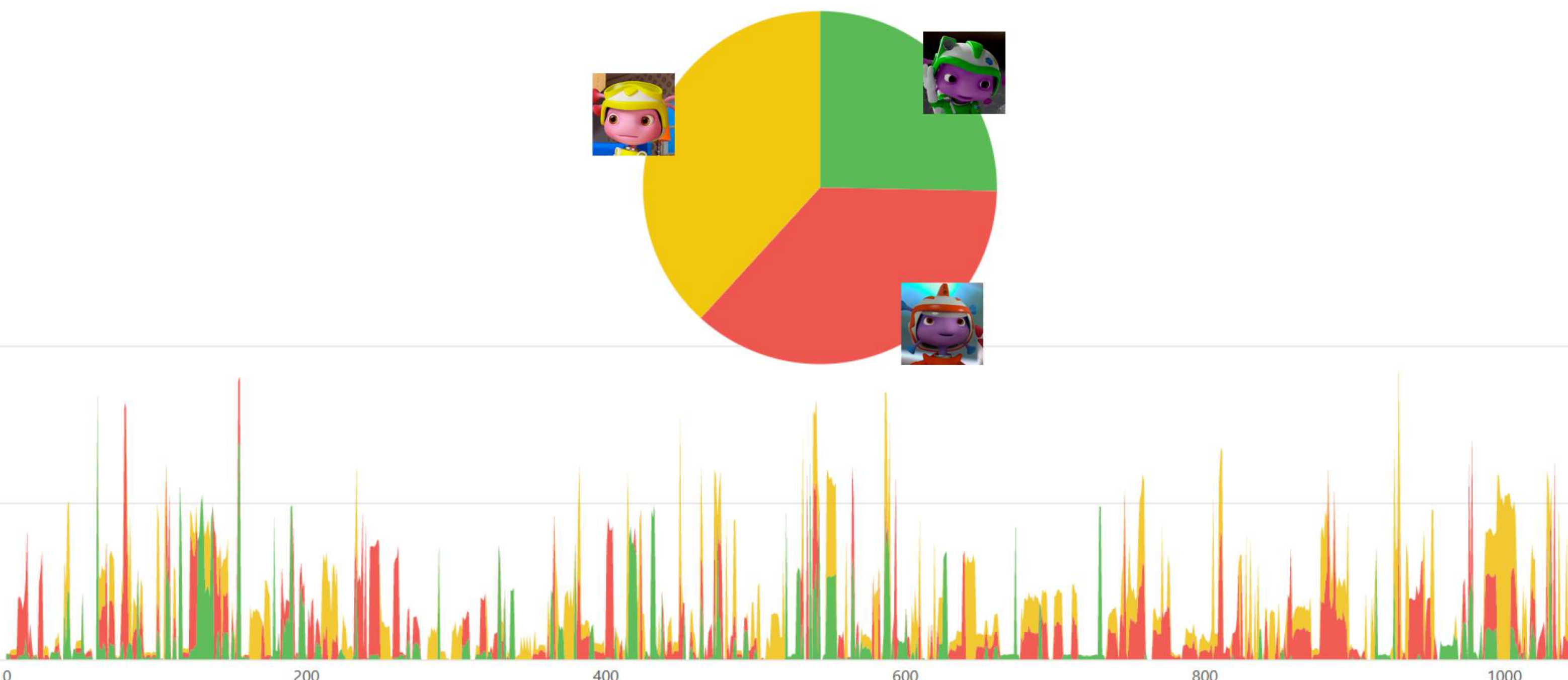


Figure 3:
Tracking the bounding boxes relative size of the three main characters in a video over time allows analysis of co-occurrences and importance. Above: a pie chart showing the aggregated occurrences over the whole video.

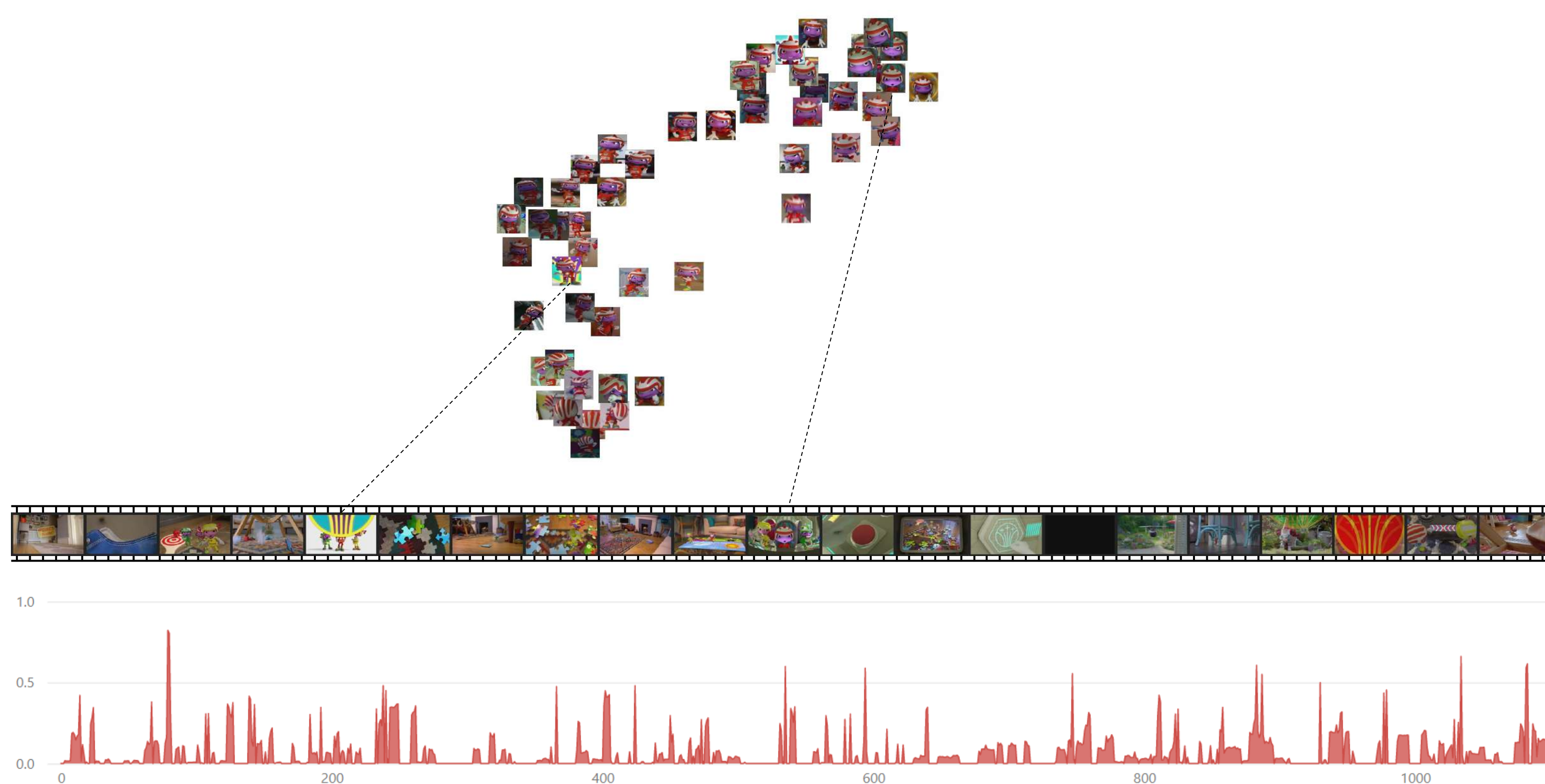


Figure 5:
A projection onto 2D space of the semantic embedding of a specific character allows to index into its depictions in the video. The lower graph shows the bounding box size of this character over time. Using such an interface, an animator can search for a character in a specific pose and reuse it.

Figure 1:
The pipeline of CAST [1] (left) - self-supervised representation learning pipeline includes: automatic shot segmentation, detection, multi-object tracking, and contrastive-based refinement to define the unique semantic embedding for an animation style. This semantic space allows to build more accurate classifiers and design better analysis and visualization tools.

References:
[1] O. Nir, G. Rapoport, and A. Shamir.
Cast: Character labeling in animation using self-supervision by tracking.
Computer Graphics Forum, 41(2):135–145, 2022.